

# Lag-Enhanced LSTM for Power Grid Frequency Prediction

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**Abstract**— This paper investigates the impact of lagged features on short-term frequency prediction using three univariate Long Short-Term Memory (LSTM) models: without lag, with lag steps 1 to 3, and with lag steps 1 to 10. The baseline model yielded a MAPE of 8.08%, which was reduced by 89.4% to 0.86% when using lag steps 1 to 3. However, extending the window to 10 lag steps (equivalent to 200 ms) slightly worsened performance, raising the MAPE to 1.08%. Correlation and delay analyses revealed that frequency synchronization across the grid is highly time-sensitive and spatially structured, with most predictive information concentrated in the immediate past. These directional differences highlight that correlation degradation is not symmetric, and the direction of interaction influences how frequency deviations propagate. Identifying such asymmetric and time-sensitive relationships is essential for refining model input design, particularly for lag-aware LSTM architectures that rely on the physical structure and timing of signal transmission. These findings suggest that compact lag windows offer the best balance between model accuracy and efficiency, aligning with the physical dynamics of the grid.

**Keywords**—frequency prediction, temporal dependencies, lagged features, power grid stability

## I. INTRODUCTION

Maintaining frequency stability in power grids is essential for ensuring reliable operation, especially as renewable energy sources become increasingly integrated and grid complexity grows. Frequency stability reflects the balance between electricity supply and demand, and significant deviations from the nominal 50 Hz can lead to equipment damage or even widespread blackouts [1], [2]. Accurate prediction of frequency fluctuations helps grid operators take proactive measures to ensure reliability [3].

The nonlinear interactions between generators, loads, and renewable sources contribute to the complexity of frequency behavior [4]. These dynamics are further influenced by spatial relationships across regions in the grid, which require analytical models capable of capturing such interdependencies [5]. High-resolution time-series data has proven especially valuable for detecting emerging instabilities and informing prediction models [6].

Various methods have been proposed to forecast grid frequency, ranging from physics-based simulation models to data-driven learning approaches. While physics-based models provide interpretability, they often struggle to scale to large interconnected systems [7], [8]. In contrast, data-driven models, especially those based on deep learning, have shown strong performance in modeling nonlinear and nonstationary grid behaviors [9].

Traditional machine learning methods such as Support Vector Regression, Random Forests, and Gradient Boosting have also been widely applied for prediction tasks in power systems. SVR is effective for modeling nonlinear

relationships [10], Random Forests offer robustness and interpretability [11], and XGBoost has achieved excellent performance in short-term electric load forecasting [12]. However, these models typically process observations independently and lack the temporal memory needed to capture long-term dependencies. In contrast, Long Short-Term Memory (LSTM) networks are specifically designed to learn from sequential patterns and dynamic behaviors, making them particularly suitable for modeling frequency fluctuations in real-time grid environments. Their effectiveness has been demonstrated in grid frequency prediction tasks [13], short-term load forecasting using hybrid Convolutional Neural Network (CNN) – Long Short-Term Memory (LSTM) models [14], and spatiotemporal modeling of campus-level frequency data [15].

Although LSTM has been used in previous studies, key methodological gaps remain. For example, Khaleefah et al. [16] applied LSTM to predict smart grid stability but used simulated data with limited system size and did not describe input features or exploit temporal lag structures. Sathe and Yang [15] used ConvLSTM for frequency prediction based on power consumption in a campus microgrid, but did not consider lagged frequency features or inter-location influence.

In this study, we introduce a lag-enhanced LSTM model that explicitly incorporates frequency deviations from  $t-1$  to  $t-10$  across 14 high-voltage substations, equipped with phasor measurement devices, in the Hungarian power grid. By selecting features based on inter-station cross-correlation analysis, the model learns both how frequency disturbances propagate over time and how they travel spatially across the grid. The dataset includes over 84 million real-time data points recorded within a 24-hour window, providing a uniquely high-resolution view of short-term frequency dynamics.

This approach not only improves prediction accuracy but also enhances interpretability by revealing grid synchronization behavior and temporal delay patterns between stations. In doing so, our model bridges the gap between traditional forecasting methods and physics-informed analyses of interconnected grid dynamics. The remainder of the paper is organized as follows: Section II describes the dataset and preprocessing steps, Section III presents the methodology and experimental setup, Section IV discusses the results, and Section V concludes the paper.

## II. METHODS AND DATA

This section explains how the frequency data from multiple grid stations was prepared for short-term prediction. It covers data cleaning, filling in missing values, and aligning the time series across stations for lag analysis.

### A. Dataset and Preprocessing

This study uses a 24-hour dataset of high-frequency measurements from 14 substations in the Hungarian

transmission grid. The dataset contains approximately 85 million entries, with each value representing the deviation from the nominal 50 Hz frequency at a specific time and location. These short-term fluctuations, recorded every 20 milliseconds, allow the model to learn how frequency behaves across different parts of the grid.

For modeling purposes, only three fields were retained: station name, measurement timestamp, and frequency oscillation value. The data was collected through a national infrastructure operated by the Hungarian transmission system operator (TSO) MAVIR, with geographically distributed measurement points that allow high-resolution monitoring of frequency variations. Although this study uses a 24-hour sample, the data structure and acquisition process are consistent with the measurement setup described by Hartmann et al. [17].

In this research, preprocessing involved two main steps: handling missing values and aligning data across stations, implemented using Python with pandas[18] and NumPy[19]. Frequency data from each station was cleaned individually. Missing values were filled with the station's own mean to preserve its characteristics [20]. Stations with only missing data or zero variance were excluded, and all values were converted to a consistent numeric format.

To analyze lagged relationships such as  $t - 1$  or  $t - 2$ , we shifted each station's time series by a fixed number of rows, where each row corresponds to a 20-millisecond interval. For example, a lag  $-3$  represents a 60 ms forward shift. This approach allowed us to model short-term dependencies across stations at multiple temporal offsets. All stations contributed both as references and lagged targets, forming a fully connected structure that captures how frequency changes propagate across the grid.

### B. Lag-Based Correlation Setup

We computed pairwise cross-correlations across lag intervals from  $t - 1$  to  $t - 10$  for all station combinations. Several station pairs showed correlation values consistently above 0.997 across multiple lag intervals, indicating strong temporal relationships between frequency patterns at different grid locations. Each station contributed to the correlation process both as a reference point at time  $t$  and as a target at shifted time steps.

The Pearson correlation coefficient used to quantify the strength of the lagged relationships is defined as [21]:

$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \quad (1)$$

where

$x_i$  and  $y_i$  represent frequency values of two stations at aligned time steps,  $\bar{x}$  and  $\bar{y}$  are their means,  $r_{xy}$  is the correlation coefficient.

Fig. 1 presents a detailed temporal comparison between two stations, where the frequency signal from Station A is shifted by one time step and overlaid with the original signal from Station B. The lower subplot shows the two frequency traces, revealing a high degree of alignment in both trend and amplitude across the observed interval. The similarity in slope and turning points suggests that frequency deviations at Station A may precede and closely resemble those at Station

B, supporting the hypothesis of a short-range dynamic propagation.

To quantify this alignment, Fig. 2 shows the absolute difference between the two signals over the same period. Most differences remain below 2.0, with a majority clustering between 1.2 and 1.8. Spikes beyond 2.5 are infrequent and isolated, indicating that the lag-1 relationship between these stations remains consistent for most of the interval. These results suggest that lagged frequency inputs from neighboring stations may capture meaningful predictive signals [22], especially when deviations propagate rapidly through the grid [23]. This behavior forms the basis for the correlation and delay analysis presented in subsequent sections.

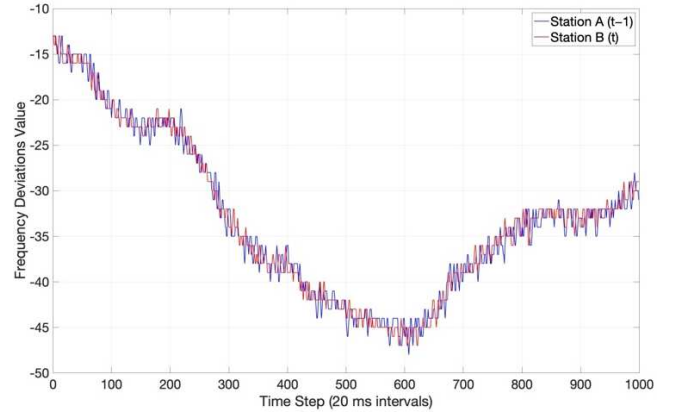


Fig. 1. Frequency Comparison Between Station A ( $t-1$ ) and Station B ( $t$ )

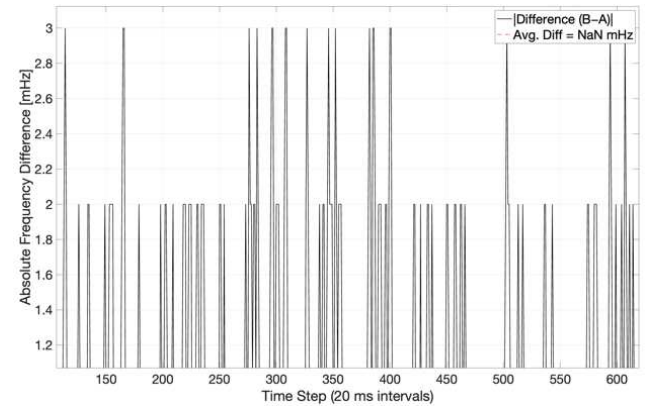


Fig. 2. Absolute Difference Between Station A ( $t-1$ ) and Station B ( $t$ )

### C. Normalization

The Min-Max scaling was applied to normalize all frequency values between 0 and 1, enhancing model performance and convergence during training.

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (2)$$

where [24]  $x'$  is the normalized value,  $x_{min}$  and  $x_{max}$  are the minimum and maximum frequency values.

### D. LSTM Model Design and Implementation

The following outlines the LSTM model architecture, designed to utilize sequential patterns and temporal dependencies for frequency prediction.

#### 1. Input Layer:

- Lagged Features Input: Processes Lag 1, Lag 2, ..., Lag 10 features from individual selected stations.

## 2. LSTM Layers:

- First LSTM layer with 50 units, focusing on extracting temporal dependencies.
- Second LSTM layer with 50 units, refining the temporal sequence information.
- Dropout layer to prevent overfitting.

## 3. Dense Layers:

- Fully connected (Dense) layer with ReLU activation for nonlinearity.
- Output layer with a single neuron for frequency prediction.

## 4. Training Configuration:

- 80% of the data was allocated for training, and 20% for testing.
- The model employed the Adam optimizer, with a learning rate of 0.001, and ran for 10 epochs with a batch size of 32.

LSTM Cell Formula [25]:

$$h_t = \sigma(W_x x_t + W_h h_{t-1} + b_h) \quad (3)$$

where  $h_t$  is the hidden state at time  $t$ ,  $x_t$  is the input vector,  $W_x$  and  $W_h$  are the weight matrices for the input and hidden state, respectively,  $b_h$  is the bias term,  $\sigma$  is the activation function.

## III. RESULTS AND ANALYSIS

This section presents the main results, including correlation patterns, delay behavior, and prediction accuracy using different lag inputs.

### A. Correlation Structure at Multiple Lags

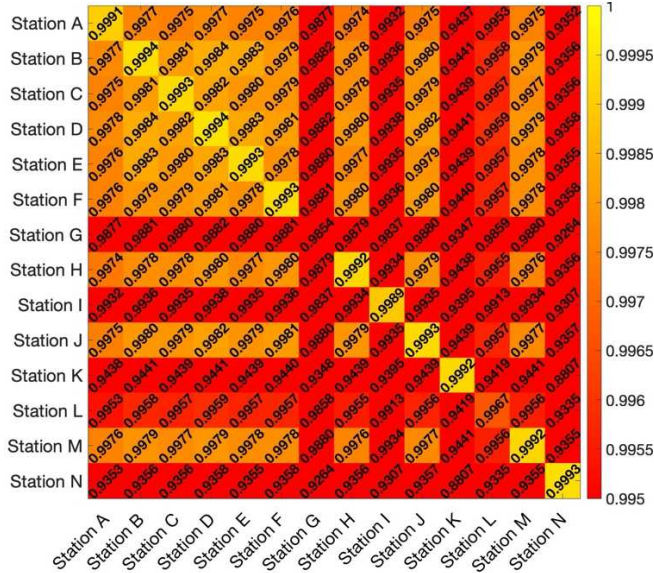


Fig. 3. Cross-Correlation Matrix for Lag-1

At lag  $t-1$  (Fig. 3) the cross-correlation matrix shows that most station pairs are highly synchronized, with values above 0.997 and several even exceeding 0.999. For example, Station C to D reaches 0.9996, and H to I is 0.999, supporting the hypothesis in Section I that frequency changes spread very quickly across closely connected areas. In contrast, Station G consistently shows lower correlations, such as 0.9854 to F and 0.9861 to C, suggesting it responds differently, potentially due

to factors such as network topology or local operating conditions, which are not further examined in this study. At lag  $t-10$  (Fig. 4) or 200 milliseconds, some connections like C to D remain strong at 0.9995, but others begin to weaken. For instance, L to H remains high at 0.9991, while the reverse direction is only 0.9942, showing that influence between stations is not always equal. These differences highlight how timing affects signal relationships and why lag-specific inputs, as described in Section II, are necessary for reliable predictions.

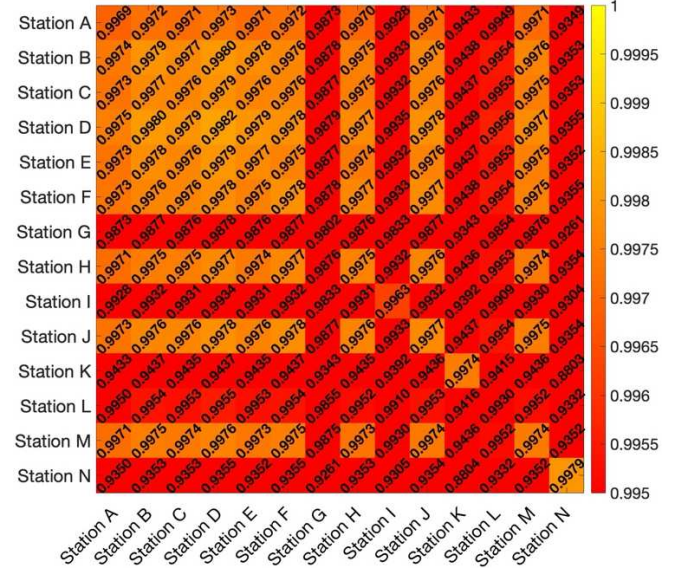


Fig. 4. Cross-Correlation Matrix for Lag-10

Besides this general weakening over time, some station pairs drop much more noticeably. Station N, for example, falls to 0.9264 toward D and 0.9307 to E at lag  $t-10$ , making it one of the least synchronized stations. Similarly, Station K to D drops from 0.9441 at lag  $t-1$  to 0.9406, and Station G to C drops from 0.9861 to 0.9833. These results contrast with stable pairs like C to D and show that some stations, such as N or G, have lower and less stable correlations at lag  $t-10$ . This suggests that their frequency patterns are less closely aligned with other stations as time passes. This suggests that their frequency patterns are less coupled with those of other stations, indicating weaker dynamic interaction within the system.

Note that the diagonal values in the cross-correlation matrices (Fig. 3 and Fig. 4) are not exactly 1.0000. This is because they represent correlations between a station's own frequency at time  $t-k$  and at time  $t$ , not at the same time step. In other words, these are lagged self-correlations, which can naturally fall below 1 due to short-term fluctuations in the signal. Small differences are expected and reflect the dynamic nature of frequency behavior in real grid conditions.

Fig. 5 presents a symmetrical heatmap combining the maximum lagged cross-correlation between all station pairs and the corresponding delay in milliseconds at which this peak occurs. Most station pairs exhibit very high correlations above 0.990, with many reaching 0.998 or higher. The delays are predominantly concentrated at the first lag step (20 milliseconds), indicating that many nodes respond to each other almost immediately within a single measurement interval. Such fast and strong coupling supports the assumption in Section I that frequency disturbances propagate rapidly and coherently across the network. From a physical

perspective, these minimal delays suggest direct influence through strong electrical paths and methodologically validate the use of short lags in the LSTM input structure as described in Section II.

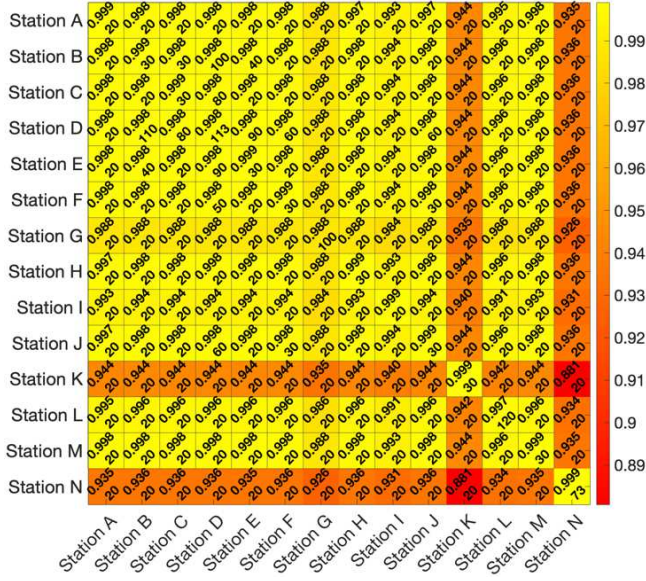


Fig. 5. Maximum Correlation and Delay Across All Station Pairs

Several deviations are noteworthy. Station N reaches its highest correlation with itself at 40 milliseconds instead of 20, suggesting unstable or delayed internal dynamics. Station K shows its best correlation with other nodes such as Station M and Station L only at 60 milliseconds, and the values are relatively low, around 0.910 to 0.925. Station M also interacts weakly with multiple others, such as 0.895 with Station N, despite being part of the same grid. These observations imply reduced dynamic influence and slower response, possibly due to topological distance, control delays from inverter-based generation, or localized disturbances. Not all high correlations imply fast propagation, the pair Station J and Station L reaches 0.996 at 40 milliseconds, showing that strong coupling can still involve time lag. This confirms that both the magnitude and timing of interaction matter and highlights why the model must treat lag not as a numerical artifact but as a meaningful physical signal.

### B. Temporal and Spatial Patterns of Delay

To understand how stations organize based on delay behavior, hierarchical clustering was applied to the delay matrix. Fig. 6 presents a hierarchical clustering dendrogram based on the delay matrix, where the vertical axis shows the linkage distance between stations. The clustering is derived from average delays at which each station pair reached maximum correlation, reflecting how similarly each node responds to or influences others over time. Three clusters were formed at a linkage threshold of 50 ms.

Cluster 1 contains the majority of stations including A, I, M, K, F, and others, indicating these nodes share similar temporal behavior and respond to frequency changes within comparable timeframes. Their grouping suggests that they are part of a tightly connected region in the grid where disturbances propagate with minimal delay.

The clustering result in Fig. 6 shows that Station D and Station L are separated from the main group because of how their delay values differ from other stations. Station D has

higher delays to most stations, often around 40 milliseconds, while most other stations are connected with shorter delays of about 20 milliseconds. Station L is different in another way. It has short delays to most stations, but to a few specific stations like J, K, M, and N the delays are much longer, reaching up to 73 milliseconds. This unusual combination makes its delay pattern different from the others. As a result, both D and L appear in separate clusters. This shows that clustering depends not just on the average delay but on how each station connects to all others in terms of timing.

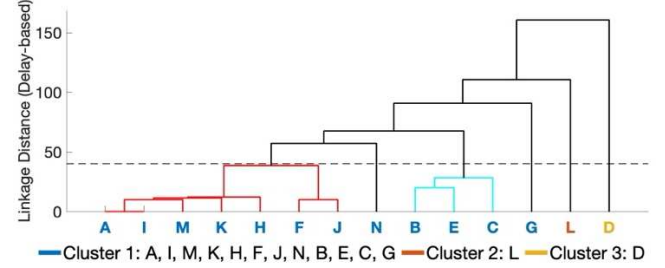


Fig. 6. Hierarchical Clustering Based on Delay Behavior

In real-world power systems, each cluster shows how similar the stations are in terms of delay and response. Cluster 1 contains stations that respond quickly and behave in a similar way. Cluster 2 and Cluster 3 include stations with slower or different response patterns. This helps identify which stations follow the main system behavior and which ones may need to be handled separately.

Fig. 7 shows the correlation trajectories across lag-1 to lag-10 for selected station pairs with peak correlation values above 0.9982. Among them, two pairs stand out with a notably sharp decline: Station B to Station D and Station C to Station D. Both start above 0.9985 at lag-1 but drop below 0.9975 by lag-10. For example, the correlation from Station C to D decreases from 0.9986 at lag-1 to 0.9973 at lag-10, marking a decline of over 0.0013. This suggests that Station D only maintains strong coupling with B and C within a narrow delay window and rapidly loses temporal alignment as lag increases. The steep decrease indicates that Station D responds quickly to nearby nodes but becomes decoupled over time, potentially due to its location on the grid, limited feedback mechanisms, or local control behavior.

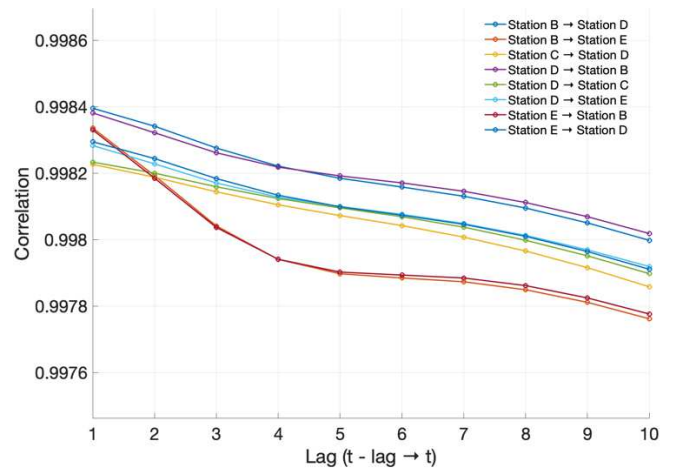


Fig. 7. Correlation Decay from Lag-1 to Lag-10 for Selected Pairs

Other pairs reveal contrasting temporal patterns. The correlation from Station E to Station D decreases modestly

from 0.9985 at lag-1 to 0.9978 at lag-10, showing that the influence from Station E remains relatively stable across time. Interestingly, the reverse direction from D to E declines even more gradually, from 0.9984 to 0.9981, suggesting that Station D has a longer-lasting impact on Station E than vice versa. A similar asymmetry appears in Station D to Station C, which falls from 0.9985 to 0.9980, while C to D drops more steeply. These directional differences highlight that correlation degradation is not symmetric, and the direction of interaction influences how frequency deviations propagate. Identifying such asymmetric and time-sensitive relationships is essential for refining model input design, particularly for lag-aware LSTM architectures that rely on the physical structure and timing of signal transmission. Overall, the varying rates of correlation decay across directions and station pairs reveal distinct temporal roles within the grid, distinguishing fast-responding nodes from those that serve as delayed propagators or dynamic endpoints.

Fig. 8 shows the spatial distribution of delay in frequency synchronization across the Hungarian grid, using Station A as the reference point. The color contour represents the time each station takes to align with frequency changes originating from Station A, with warmer colors indicating shorter delays and cooler tones indicating longer ones. Stations L, G, and I, located in the central and northeastern region, show the shortest delays at 20 to 40 milliseconds, indicating strong synchronization with the source. In contrast, Stations D, H, and J in the southwestern part of the grid exhibit longer delays up to 80 milliseconds, reflecting slower alignment with the reference signal.

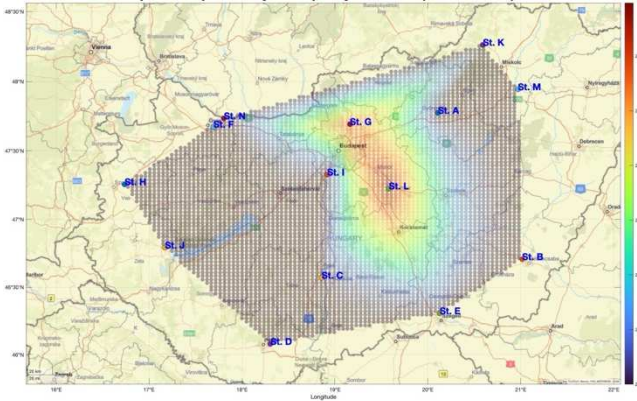


Fig. 8. Spatial-temporal Delay of Frequency Oscillation (from Station A)

The delay gradient clearly follows a spatial path from the northeast to the southwest, suggesting that frequency disturbances originating from Station A propagate most rapidly toward the central and eastern zones, while western and southern stations respond more slowly. If a frequency event occurs near Station A, the central corridor including Stations L and I will be affected almost immediately, while the southwestern region, including Stations D, H, and J, may experience delayed synchronization and temporary phase mismatch. This spatial gradient illustrates how frequency disturbances propagate through the network and reveals the dynamic structure of the grid in real time. Rather than treating all nodes as equally connected, this map highlights the directional nature of synchronization, shaped by electrical distance, generator response characteristics, and network topology. Such spatial delay information offers practical insight into how different areas of the grid contribute to or fall behind in maintaining coherent frequency behavior.

### C. LSTM Prediction with and without Lagged Features

To evaluate the effect of temporal information on frequency prediction, three LSTM models were developed with different input strategies. Model training was conducted in Python (TensorFlow 2.11, Keras), with preprocessing in Pandas and NumPy, and evaluation using Scikit-learn. The first model uses only the current frequency value from a single station (univariate input without lag), assuming that present information alone is sufficient for forecasting. The second model incorporates lagged values from lag-1 to lag-3 to capture recent memory effects, while the third model extends this window to lag-1 through lag-10 in order to account for longer term temporal dependencies. All models share the same architecture: two LSTM layers followed by a dense output layer that produces a single step forecast. This design was selected to isolate the effect of temporal depth while maintaining a consistent model structure for fair comparison.

Fig. 9 shows the error comparison between the three models. In this case, error means the difference between the predicted and actual frequency values, measured by MAE, MSE, RMSE, and MAPE. The model without lag gives the highest MAPE at 8.08 percent, which shows that using only current values is not enough to capture how frequency changes over time. Adding short-term memory with lag 1 to 3 improves the result significantly, lowering the MAPE to 0.86 percent, or an improvement of 89.4 percent. This shows that recent data gives most of the useful information, helping the model predict short-term changes more accurately.

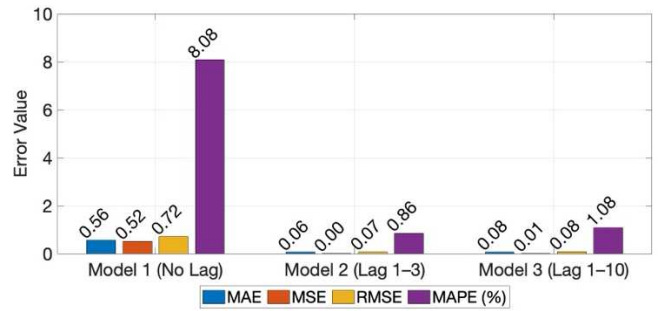


Fig. 9. Comparison of Evaluation Metrics by LSTM Models

When the lag window is extended to lag-1 to lag-10, MAPE rises slightly to 1.08%, a 25.6% deterioration compared to the lag 1-3 model. Although still much better than the baseline, this drop suggests that distant lags add little useful information and may introduce noise or weaken the impact of key features. In LSTM training, longer sequences raise the risk of vanishing gradients and can lead the model to learn less meaningful temporal patterns. This highlights the need to align lag design with the system's dominant time scale.

Fig. 10 shows the prediction results for the first 20 time steps at Station A, comparing actual frequency values with three LSTM models: one without lag, one using lag-1 to lag-3, and one using lag-1 to lag-10. The model without lag fails to capture key signal features. It underestimates the peak at steps 5 to 6, misses the drop at steps 9 to 13, and generally produces a flattened response, showing that it cannot learn system dynamics from current values alone.

The lag-1 to lag-3 model closely follows the actual signal, capturing the rise to 4.0, the drop below 2.0, and the recovery, with only minor deviations. The lag-1 to lag-10 model captures the trend but lags in transitions and adds spurious

oscillations after step 13. This likely results from longer input interacting with LSTM's sequential structure, where more lagged values can lead to overfitting on less important patterns. As predictions continue step by step, small errors build up and affect alignment, especially in later steps.

These differences confirm that short lag windows give the best balance between responsiveness and stability, as also shown in the metrics. MAPE drops from 8.08 percent (no lag) to 0.86 percent (lag-3), then slightly increases to 1.08 percent (lag-10). This shows that local frequency patterns can be predicted well with minimal input history, while longer lag depth may introduce noise and reduce precision.

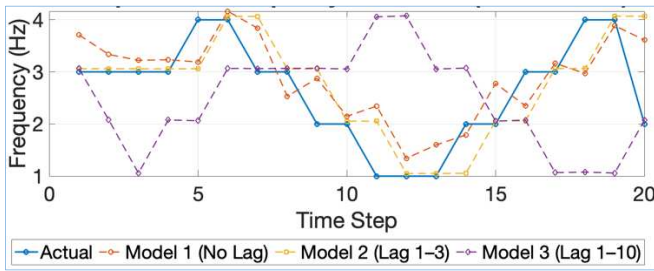


Fig. 10. Predicted vs Actual Frequency Deviation Over 20 Time Steps (Station A)

#### IV. CONCLUSION

This study demonstrates the benefits of incorporating short-term lagged features in improving the accuracy of frequency prediction in power systems. The baseline model without lag yielded the highest prediction error, with MAE = 0.5577 and MAPE = 8.08%, indicating poor responsiveness to dynamic changes. Introducing lagged inputs from lag-1 to lag-3 drastically improved the results, reducing the MAE to 0.0626 and MAPE to 0.86%, which represents a reduction of 88.8% and 89.4% respectively. Surprisingly, expanding the lag window to lag-1 to lag-10 slightly degraded performance, with MAE increasing to 0.0751 and MAPE to 1.08%, suggesting that the most predictive temporal information is concentrated within a narrow recent window. These findings confirm that frequency behavior in the Hungarian power grid is driven by short-term temporal patterns and that predictive models benefit most from capturing these immediate dependencies, while future research may enhance model robustness and generalizability by integrating physical grid features such as electrical distance or impedance and extending evaluation to multi-station and variable operating conditions. This study focuses on point prediction using LSTM. Incorporating uncertainty estimation, such as confidence intervals or probabilistic forecasting, would enhance the reliability and applicability of the results and is suggested for future work.

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